# The Poisson Algebra of Pairwise Interactions: A Calogero-Moser Integrability Test

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The Poisson algebra generated by the pairwise Hamiltonians of a three-body system carries a natural filtration whose dimension sequence $d(k)$ is a structural invariant. For Newtonian gravity and all other singular pair potentials tested to date, this sequence takes the universal value $[3, 6, 17, 116, \ldots]$, while the harmonic oscillator gives a finite algebra with $d(k) = 15$ for $k \geq 3$. We test whether this dichotomy reflects integrability by computing $d(k)$ for the one-dimensional Calogero-Moser system with $1/r^2$ interactions—an exactly integrable model that arises naturally in Galperin's billiard construction for the digits of $\pi$. We find $d(k) = [3, 6, 17, 116]$ for the integrable Calogero-Moser model, identical to non-integrable gravity, and invariant across all Galperin superintegrable mass ratios. The Poisson algebra dimension sequence therefore detects the *singularity class* of the pair potential—not its integrability properties.

## INTRODUCTION

A remarkable observation connects two seemingly unrelated problems in mathematical physics. Galperin [1] showed that counting collisions between two balls and a wall—a classical one-dimensional three-body system—yields the digits of $\pi$ when the mass ratio takes the form $m/M = 10^{2N}$. Davis, Geyer, and Ravishanker [2] extended this analysis, demonstrating that the billiard dynamics map exactly onto the Calogero-Moser model with inverse-square interactions. In particular, the squared angular momentum $L^2 = (M + m)^2(Yw - yW)^2$ is a conserved quantity, rendering the system Liouville integrable, while at special mass ratios $m/M = \tan^2(\pi/q)$ for integer $q$, a third independent integral (the Chevalley invariant $J$) appears, yielding superintegrability.

Independently, recent investigations of the three-body problem have revealed that the Poisson algebra $\mathcal{A}$ generated by the three pairwise Hamiltonians $\{H_{12}, H_{13}, H_{23}\}$ carries a rich algebraic structure. Using auxiliary variables $u_{ij} = 1/r_{ij}$ to maintain polynomial expressions, the iterated Poisson brackets of these generators can be organized into a filtration $\mathcal{A}_0 \subset \mathcal{A}_1 \subset \mathcal{A}_2 \subset \cdots$, where $\mathcal{A}_k$ is spanned by brackets of depth at most $k$. The dimension sequence $d(k) = \dim \mathcal{A}_k$ is a computable structural invariant.

For Newtonian gravity ($V \propto 1/r$) with $N = 3$ bodies, this sequence is $d(k) = [3, 6, 17, 116, \ldots]$, and it is invariant under changes in spatial dimension ($d = 1, 2, 3$), mass ratios, and coupling constants. Strikingly, *every* singular pair potential tested—$1/r^2$, $1/r^3$, $\log r$, Yukawa, and Coulomb—produces the identical sequence. In contrast, the harmonic oscillator ($V \propto r^2$) generates a finite algebra with $d(k) = [3, 6, 13, 15, 15]$, saturating by level 3.

This dichotomy raises a natural question: does the Poisson algebra dimension sequence distinguish integrable from non-integrable systems? The Calogero-Moser model in one dimension is the ideal test case—it is exactly integrable, with known conserved quantities, yet its pair potential $V \propto 1/r^2$ is singular. If integrability constrains the growth of the pairwise Poisson algebra, the one-dimensional Calogero system should show a dimension sequence different from non-integrable gravity. If it does not, the sequence reflects something else entirely: the singularity structure of the interaction.

In this paper, we report the computation of $d(k)$ for the one-dimensional Calogero-Moser three-body system through filtration level $k = 3$. We find $d(k) = [3, 6, 17, 116]$—identical to every other singular potential. The result is invariant across all Galperin superintegrable mass ratios ($q = 3, 4, 5, 6$) and generic non-superintegrable mass configurations.

## METHOD

### Pairwise Hamiltonians and Auxiliary Variables

Consider $N = 3$ particles in $d$ spatial dimensions with positions $\boldsymbol{q}_i$ and momenta $\boldsymbol{p}_i$ ($i = 1, 2, 3$), interacting through a pairwise potential $V(r_{ij})$. The three pairwise Hamiltonians are

$$H_{ij} = T_i + T_j + V(r_{ij}), \qquad T_i = \frac{|\boldsymbol{p}_i|^2}{2m_i}, \tag{1}$$

where $r_{ij} = |\boldsymbol{q}_i - \boldsymbol{q}_j|$.

For singular potentials $V \propto r_{ij}^{-n}$, Poisson brackets of the $H_{ij}$ generate expressions involving inverse powers of $r_{ij}$ and derivatives thereof. To maintain polynomial structure, we introduce auxiliary variables $u_{ij} = 1/r_{ij}$ and augment the phase space with the chain rule relations

$$\frac{\partial u_{ij}}{\partial q_i^\alpha} = -u_{ij}^3 \, (q_i^\alpha - q_j^\alpha), \tag{2}$$

where $\alpha$ indexes spatial components. In terms of these variables, $V_{ij} = -g \, m_i m_j \, u_{ij}^n$ and all Poisson brackets reduce to polynomial expressions in $(\boldsymbol{q}, \boldsymbol{p}, \boldsymbol{u})$.

### Filtration and Dimension Sequence

The *filtration level* of a generator is defined recursively: $H_{ij}$ has level 0, and $\{f, g\}$ has level $\ell(f) + \ell(g) + 1$. We define

$$\mathcal{A}_k = \mathrm{span}\{g : \ell(g) \leq k\}, \qquad d(k) = \dim \mathcal{A}_k. \tag{3}$$

The dimension sequence $d(0), d(1), d(2), \ldots$ is a structural invariant of the algebra.

### Computational Procedure

At each level $k$, we compute all Poisson brackets between generators at levels $k_1$ and $k_2$ with $k_1 + k_2 + 1 = k$, simplify the resulting expressions via `sympy.expand`, and collect them as candidate new generators. The functional independence of all generators through level $k$ is determined by constructing an evaluation matrix

$$M_{si} = g_i(\boldsymbol{z}_s), \qquad s = 1, \ldots, N_{\text{samples}}, \tag{4}$$

where $\{\boldsymbol{z}_s\}$ are random phase-space points drawn uniformly from a compact domain. The dimension $d(k) = \mathrm{rank}(M)$ is determined by singular value decomposition (SVD), with the rank identified by a gap in the singular value spectrum exceeding $10^7$ between adjacent values. This numerical rank determination has been validated against exact symbolic rank computation (via echelon form over $\mathbb{Q}$) in all cases where the latter is tractable.

For the Calogero-Moser potential $V_{ij} = -g\, m_i m_j / r_{ij}^2$, the Hamiltonian in auxiliary variables is $H_{ij} = T_i + T_j - g\, m_i m_j\, u_{ij}^2$. The chain rule table is unchanged because it depends only on the definition $u_{ij} = 1/r_{ij}$.

All computations were performed using SymPy for symbolic algebra and NumPy/SciPy for numerical SVD, with $N_{\text{samples}} = 500$ phase-space evaluation points.

## RESULTS

### The Critical Experiment: 1D Calogero-Moser

The one-dimensional Calogero-Moser system with $N = 3$ equal-mass particles and $V_{ij} = -1/r_{ij}^2$ is exactly integrable. Its Poisson algebra dimension sequence through level 3 is

$$d(k) = [3, 6, 17, 116]. \tag{5}$$

This is *identical* to the sequence obtained for one-dimensional Newtonian gravity ($V = -1/r$), which is not integrable. The SVD gap ratios exceed $10^7$ at each level, confirming the rank determination is unambiguous (Table I).

### Complete Potential Comparison

Table II summarizes the dimension sequence across all tested potential types and spatial dimensions. Every singular potential—$1/r$, $1/r^2$, and $1/r^3$—produces the identical sequence $[3, 6, 17, 116]$ in both one and two spatial dimensions. Prior results for $\log r$, Yukawa ($e^{-\mu r}/r$), and Coulomb potentials in two dimensions also yield this sequence. The harmonic oscillator ($V \propto r^2$) is the sole exception, generating a finite algebra that saturates at $d(3) = 15$.

### Mass Ratio Invariance at Superintegrable Points

Galperin's billiard system becomes superintegrable when the mass ratio takes the form $m/M = \tan^2(\pi/q)$ for integer $q$ [2]. At these special values, the system possesses a third independent integral—the Chevalley invariant—beyond the energy and angular momentum.

Table III shows that the dimension sequence is invariant across all tested mass ratios, including both superintegrable values ($q = 3, 4, 5, 6$) and a generic non-superintegrable configuration. In every case, $d(k) = [3, 6, 17, 116]$.

## DISCUSSION

### The Singularity Dichotomy

Our results establish a clear classification principle: the Poisson algebra dimension sequence of a three-body system with pairwise interactions depends on the *singularity class* of the pair potential, not on its integrability properties.

- **Singular potentials** ($V \sim r^{-n}$, $\log r$, Yukawa): The algebra is infinite-dimensional with the universal growth sequence $d(k) = [3, 6, 17, 116, \ldots]$. This holds regardless of the exponent $n$, spatial dimension $d$, particle masses, or coupling constants.
- **Regular potentials** ($V \sim r^2$): The algebra is finite-dimensional, saturating at $d = 15$ by level 3.

TABLE I. SVD gap ratios for the 1D Calogero-Moser ($1/r^2$) system at each filtration level.

| Level $k$ | $d(k)$ | Gap ratio | $\sigma_{d(k)}/\sigma_{d(k)+1}$ |
|---|---|---|---|
| 0 | 3 | $\infty$ | rank 3 of 3 |
| 1 | 6 | $\infty$ | rank 6 of 6 |
| 2 | 17 | $2.7 \times 10^{14}$ | $1.4 \times 10^{-1}$ / $5.3 \times 10^{-16}$ |
| 3 | 116 | $6.9 \times 10^{7}$ | $3.0 \times 10^{-4}$ / $4.3 \times 10^{-12}$ |

TABLE II. Poisson algebra dimension sequences for various pair potentials and spatial dimensions. All $N = 3$, equal masses.

| $d$ | Potential | Integ. | $d(0)$ | $d(1)$ | $d(2)$ | $d(3)$ |
|---|---|---|---|---|---|---|
| 1 | $1/r$ | No | 3 | 6 | 17 | 116 |
| 1 | $1/r^2$ | Yes | 3 | 6 | 17 | 116 |
| 1 | $1/r^3$ | No | 3 | 6 | 17 | 116 |
| 2 | $1/r$ | No | 3 | 6 | 17 | 116 |
| 2 | $1/r^2$ | No$^\dagger$ | 3 | 6 | 17 | 116 |
| 2 | $\log r$ | No | 3 | 6 | 17 | 116 |
| 2 | Yukawa | No | 3 | 6 | 17 | 116 |
| 2 | $r^2$ | Yes | 3 | 6 | 13 | 15 |

[a] $^\dagger$The $1/r^2$ potential is integrable in 1D (Calogero-Moser) but not in 2D for scalar particles.

TABLE III. 1D Calogero-Moser dimension sequences at Galperin superintegrable mass ratios and generic masses.

| $m/M$ | $q$ | SI? | $d(0)$ | $d(1)$ | $d(2)$ | $d(3)$ |
|---|---|---|---|---|---|---|
| 3 | 3 | Yes | 3 | 6 | 17 | 116 |
| 1 | 4 | Yes | 3 | 6 | 17 | 116 |
| $\approx 0.53$ | 5 | Yes | 3 | 6 | 17 | 116 |
| $1/3$ | 6 | Yes | 3 | 6 | 17 | 116 |
| $(1, 2.7, 0.4)$ | — | No | 3 | 6 | 17 | 116 |

This dichotomy is robust: the transition occurs precisely at the boundary between potentials with an essential singularity at $r = 0$ (in the variable $u = 1/r$, these are polynomial in $u$) and those that are analytic in the physical coordinates. The harmonic potential $V = r^2$ requires no auxiliary variables and generates a finite algebra because its Poisson brackets close after finitely many iterations.

### Why Integrability is Invisible

The Calogero-Moser model possesses two functionally independent conserved quantities beyond the total energy: the squared angular momentum $L^2$ (ensuring Liouville integrability) and, at superintegrable mass ratios, the Chevalley invariant $J$ (ensuring maximal superintegrability). Why does the pairwise Poisson algebra fail to "see" these integrals?

The answer lies in the nature of these conserved quantities. The angular momentum $L^2 = (M+m)^2(Yw-yW)^2$ is a *global* function of the full phase space—it does not decompose into pairwise contributions. The pairwise Hamiltonians $H_{ij}$ generate an algebra that respects the combinatorial structure of the interaction graph (which pairs of bodies interact), but does not capture collective conservation laws that emerge from the geometry of the full configuration space.

In the language of algebraic geometry, the pairwise Poisson algebra is generated by a specific *subvariety* of the full algebra of observables, one that is determined by the pair structure of the Hamiltonian. The integrals of motion of the Calogero model lie outside this subvariety—they are "invisible" to the pairwise generators, even though they constrain the dynamics.

This explains the universality: the dimension sequence counts the number of algebraically independent functions that can be built from pairwise Poisson brackets, and this count depends only on the local structure of each pairwise interaction near $r_{ij} = 0$ (which determines the chain rule derivatives), not on global dynamical properties like integrability.

### Connection to Galperin Billiards

Galperin's billiard system [1] computes $\pi$ through a counting argument that relies on the conservation of $L^2$ in the equivalent Calogero model. Davis *et al.* [2] showed that this conservation law makes the system integrable and, at special mass ratios, superintegrable. Our computation confirms that these powerful conservation laws leave no imprint on the pairwise Poisson algebra: the billiard system generates the same algebraic structure as generic, non-integrable gravity.

This result is not a deficiency of our method but rather a precise characterization of what the pairwise algebra measures. It is a *topological* invariant of the singularity class, not a *dynamical* invariant of the trajectory structure.

## CONCLUSION

We have computed the Poisson algebra dimension sequence for the one-dimensional Calogero-Moser three-body system—an exactly integrable model motivated by Galperin's billiard construction—and found $d(k) = [3, 6, 17, 116]$, identical to every other singular pair potential tested. This result holds at all Galperin superintegrable mass ratios ($q = 3, 4, 5, 6$) and for generic mass configurations.

Together with the contrasting behavior of the harmonic oscillator ($d(k) \to 15$), this establishes that the Poisson algebra dimension sequence is a *singularity class invariant*: it distinguishes potentials with essential singularities from regular ones, but is blind to integrability, superintegrability, mass ratios, spatial dimension, and coupling constants.

The universal sequence $[3, 6, 17, 116, \ldots]$ for singular potentials suggests a deeper algebraic structure—perhaps related to the combinatorics of the pairwise interaction graph—that warrants further investigation. In

particular, the sequence's relationship to known integer sequences and its behavior at higher filtration levels ($k \geq 4$, where computational costs become substantial) remain open questions.

Computational resources were provided by Amazon Web Services (EC2). Symbolic computations used SymPy; numerical linear algebra used NumPy and SciPy.

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